

A Comparative Study of EEG-Only vs. EEG-ECG Feature Sets for Seizure Prediction Using a 1D Convolutional Neural Network

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Abstract

Epilepsy is one of the most common neurological disorders, affecting an estimated 50 million people worldwide, and around one-third of patients continue to experience seizures despite medication. For these patients, a reliable warning of an oncoming seizure could enable timely intervention and meaningfully improve quality of life, motivating the development of non-invasive, wearable seizure-prediction systems.

This study investigates seizure prediction from wearable physiological signals and, in particular, whether adding a single electrocardiography (ECG) channel improves prediction over behind-the-ear electroencephalography (EEG) alone. Using the SeizeIT2 dataset [35] — wearable behind-the-ear EEG and ECG recordings from 60 people with epilepsy — we frame the task as distinguishing pre-ictal windows (the 10 minutes preceding a seizure onset) from interictal windows, while excluding the seizure itself. A 1D convolutional neural network with three convolutional blocks (batch normalization, ELU activations, max pooling, and dropout) is trained directly on z-score-normalized, sliding-window signal segments, with class imbalance handled through interictal subsampling and positive class weighting. Two pipelines that differ only by the ECG channel (EEG-only vs. EEG+ECG) are compared under an identical subject-aware, chronological 60/20/20 split, alongside a RandomForest band-power baseline, using the per-subject (within-patient) AUC as the primary metric.

We find that adding ECG yields a small but consistent improvement in within-patient ranking (mean per-subject AUC $\approx 0.50 \rightarrow 0.52$, independently reproduced by the baseline), but no gain — and a slight loss — on pooled AUC and F1. Crucially, this effect is highly patient-dependent: ECG markedly helps some patients while reducing performance for others. Our main contribution is a fair, reproducible quantification of ECG’s marginal and patient-specific value for wearable seizure prediction, and the finding that a single global model cannot exploit it — pointing to personalized, per-patient modeling as the more promising direction.

1 Introduction

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Epilepsy is clinically defined as a disorder of the brain characterized by an enduring predisposition to generate epileptic seizures [17]. It is one of the most widespread neurological disorders, affecting approximately 50 to 70 million people worldwide [1], [2], [5]. From both a clinical and societal perspective, uncontrolled epilepsy can have devastating consequences for a patient’s well-being, including impaired cognitive and social functioning, increased psychological burden, and a significantly reduced quality of life [5]. In severe cases, uncontrolled epilepsy may increase the risk of seizure-related injuries and, in rare situations, premature death [5], [25]. These consequences highlight the importance of reliable seizure prediction, as timely identification of seizure-related activity may support earlier interventions and improve patient safety [25], [26].

However, accurately predicting epileptic seizures remains a considerable challenge [25], [26]. Epileptic activity can vary substantially between patients, while physiological signals such as electroencephalography (EEG) are complex, high-dimensional, and difficult to interpret reliably in practical settings [25]. Furthermore, the unpredictable nature of seizures places a considerable psychological and physical impact on patients, increasing the need for reliable early warning systems [26]. While EEG remains the primary modality for seizure monitoring and prediction [25], physiological alterations beyond brain activity may also precede seizures. In particular, changes in cardiac activity measured through electrocardiography (ECG) have been associated with seizure-related dynamics and may provide additional predictive information [26]. This motivates the exploration of multimodal approaches that combine EEG and ECG data to improve automated seizure analysis.

To address these challenges, machine learning and deep learning techniques have increasingly been applied to seizure prediction systems based on physiological signals [25]. In particular, one-dimensional convolutional neural networks (1D-CNNs) have shown promising performance for time-series data such as EEG and ECG signals and are increasingly explored for seizure prediction tasks [27], [28]. Compared to traditional machine learning approaches, 1D-CNNs can automatically learn discriminative signal characteristics from sequential physiological data, reducing reliance on extensive manual feature engineering [27]. This capability is particularly valuable in seizure analysis, where subtle temporal patterns may be difficult to identify using conventional signal processing methods [25], [27]. By capturing discriminative signal features across temporal sequences, 1D-CNNs offer potential for automated seizure prediction tasks [27].

This study investigates the application of a one-dimensional convolutional neural network (1D-CNN) for automated seizure prediction using combined physiological signal data. EEG and ECG recordings from a multimodal seizure dataset are used to investigate the feasibility of combined EEG and ECG signal modelling for seizure prediction. In addition to evaluating model performance, explainability techniques are applied to provide greater insight into which signal regions contribute most to model predictions. The objective of this research is to assess the feasibility and effectiveness of a deep learning approach for seizure prediction using combined EEG and ECG signals. The remainder of this paper is structured as follows: Section 2 reviews related literature and relevant prior research. Section 3 presents the methodology, including data preprocessing, model architecture, and explainability approach. Section 4 describes the experimental setup and reports

the obtained results. Section 5 discusses the findings, evaluates model performance, and addresses limitations. Finally, Section 6 concludes the study and outlines directions for future research.

2 Related Work

Machine learning has become an increasingly important tool for automated seizure detection and prediction using EEG recordings [6,30]. Earlier approaches commonly relied on manually engineered features extracted from EEG signals, combined with classifiers such as Support Vector Machines (SVMs), k-Nearest Neighbors (KNN), and Random Forests [29,30]. More recent studies increasingly employ deep learning methods, such as Convolutional Neural Networks (CNNs) [12,29,30], which can automatically learn representations directly from raw EEG signals, reducing the need for explicit feature engineering [29].

Recent research has explored both traditional machine learning and deep learning approaches for EEG-based seizure detection. A 2024 study compared multiple classifiers, including Random Forest, XGBoost, TabNet, and a 1D Convolutional Neural Network (CNN), using the UCI Epileptic Seizure Recognition dataset [31]. The authors reported high classification performance, with the 1D-CNN achieving an accuracy of 99%, outperforming the traditional machine learning approaches. However, the study relied on a preprocessed version of the Bonn EEG dataset rather than raw continuous EEG recordings, which may limit the generalizability of the results to real-world clinical settings. Furthermore, the study focused primarily on seizure detection rather than early seizure prediction, which remains a considerably more challenging task.

In contrast to studies relying on heavily preprocessed benchmark datasets, Truong et al. investigated seizure prediction using continuous intracranial and scalp EEG recordings across multiple datasets, including CHB-MIT [32]. Their approach employed a CNN using short-time Fourier transforms (STFT) of EEG signals as input, reducing the need for patient-specific handcrafted feature engineering common in earlier machine learning methods [32]. The model achieved a sensitivity of approximately 81% on the CHB-MIT dataset while maintaining relatively low false prediction rates [32]. Although this performance was lower than the near-perfect accuracies reported in the 2024 study discussed previously, the use of continuous EEG recordings and clinically relevant prediction horizons provides a more realistic indication of real-world applicability. Furthermore, the focus on seizure prediction rather than detection makes the approach more closely aligned with practical early-warning systems for epilepsy patients [32].

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More recently, Saeizadeh et al. proposed a multimodal deep learning framework combining non-invasive EEG and ECG signals for seizure prediction [33]. Their approach employed separate 1D Convolutional Neural Networks (1D-CNNs) for EEG and ECG signal processing, after which predictions from both modalities were combined to improve classification performance [33]. Using data from 29 epilepsy patients, the multimodal model achieved an average sensitivity of 95%, specificity of 98%, and accuracy of 97% while predicting seizures up to one hour before onset [33]. In contrast to EEG-only approaches, the study highlights the potential value of incorporating cardiac activity into seizure prediction, suggesting that EEG and ECG may provide complementary physiological information [33]. However, the relatively limited patient cohort and personalized

training setup may still pose challenges for broader clinical generalization [33].

Overall, previous research demonstrates promising results for seizure prediction using both traditional machine learning and deep learning approaches. However, differences in dataset realism, prediction objectives, and physiological modalities make direct comparison between studies challenging. While recent multimodal approaches combining EEG and ECG signals suggest improved predictive potential, limitations in generalizability and real-world applicability remain. These findings motivate further research into clinically relevant and lightweight seizure prediction systems that can operate using complementary physiological signals.

3 Methodology

The project followed a streamlined research pipeline focused on seizure prediction using physiological signals. The methodology emphasizes a robust, reproducible workflow that remains academically grounded and feasible within a 10-week project window.

3.1 Dataset and Data Preparation

This study utilizes the SeizeIT2 dataset (OpenNeuro Accession Number: ds005873, version 1.1.0), focusing on a subset comprising subjects 001 through 060. While the dataset contains 60 subjects total, the paired EEG vs. EEG+ECG comparison specifically utilizes the 56 subjects who possess an accompanying ECG recording. Unlike traditional clinical scalp EEG setups, SeizeIT2 features multi-modal data recorded via a specialized behind-the-ear wearable Sensor Dot device. This configuration yields highly localized wearable EEG signals along with synchronized electrocardiography (ECG), electromyography (EMG), and accelerometry movement data.

The raw physiological signals were extracted from Brain Imaging Data Structure (BIDS) compatible EDF files and converted into synchronized multichannel time series at a sampling rate of 256 Hz. Preprocessing included power-line noise removal via a 50 Hz notch filter, per-recording z-score normalization, and sliding-window segmentation. Continuous streams were segmented into fixed windows of 2 seconds with a 1-second step size (50% overlap).

Rather than framing the task as seizure detection, this study strictly focuses on seizure prediction. Each window was labeled as either pre-ictal (the 10-minute period immediately preceding a seizure onset) or interictal (the baseline period between seizures). The seizure event itself and a 10-minute post-ictal guard period were explicitly excluded from the dataset. A window is labeled as pre-ictal if it exhibits any temporal overlap with the pre-ictal interval $[\text{onset} - 10 \text{ min}, \text{onset})$. Labeling assignment followed a strict priority rule where excluded regions override pre-ictal labels, which in turn override interictal labels. For multi-modal integration, the raw wearable ECG signal channel was simply synchronized, concatenated, and normalized alongside the EEG channels as a third input stream, allowing the primary deep learning network to automatically learn cardiac features from the raw waveform without manual heart rate or HRV extraction. A summary of the dataset characteristics and the window-level sample distribution is provided in Table 1.

Table 1: Summary of the SeizeIT2 (Subjects 001–060) dataset and window-level samples used in this project.

Attribute	Value / Description
Dataset	SeizeIT2 Wearable Database (OpenNeuro ds005873)
Recording format	BIDS-compatible EDF files with accompanying TSV seizure annotations
Subjects / recordings used	60 subjects (sub-001 to sub-060)
EEG channels per window	Wearable behind-the-ear configuration (e.g., cross-head configurations)
ECG-derived features	Temporal heart rate, RR intervals, and heart rate variability indicators
Window length / step size	2s / 1s
Total windows (train/val/test)	1.095.773
Pre-ictal windows	185.161
Interictal windows	910.612
Input tensor shape per sample	2×512 steps / 3×512

3.2 Key Variables, Class Imbalance, and Limitations

The primary target variable is the binary window label: pre-ictal or interictal. Input variables consist of the localized behind-the-ear wearable EEG amplitudes within each sliding window and, where applicable, the raw synchronized ECG signal amplitudes. Seizure prediction labels are assigned from annotated intervals in the BIDS TSV summary files using an any-overlap rule within the 10-minute pre-ictal horizon.

The dataset exhibits a severe class imbalance at the window level because seizure events occupy only a minor fraction of the total recording hours across the cohort. This imbalance was handled via a dual-mechanism approach: first, interictal windows were heavily undersampled during pre-processing to a fixed 5:1 ratio relative to pre-ictal windows; second, positive class weighting was applied to the loss function during training to account for the remaining class imbalance. Additional limitations include patient-specific signal heterogeneity inherent to real-world wearable data, motion artifacts affecting the sensor interface, and potential differences in sensor placement across the subject cohort, which may limit generalization to standard clinical scalp setups.

3.3 Input Feature Engineering and Selection

Input features were constructed rather than selected from a pre-existing tabular feature set. For the wearable EEG, each model input is a fixed-length multichannel window of preprocessed signal amplitudes after powerline filtering and per-channel z -score normalization; no manual frequency-

band features were extracted. For the wearable ECG, heart-rate-related features were computed from the raw signal and aligned to the corresponding EEG window timestamps.

In contrast, feature engineering was explicitly conducted for the baseline configuration. A Random Forest classifier was implemented as an interpretable baseline model, operating on handcrafted frequency domain features. Specifically, the baseline extracts the log band-power across five standard EEG frequency bands: delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), and gamma (30-45 Hz) per channel, producing a compact feature vector per window. Post-hoc interpretability of the deep learning model was assessed using Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize which temporal signal regions contribute most to predictions.

3.4 Dimensionality Reduction

No separate dimensionality reduction step (such as Principal Component Analysis or t-Distributed Stochastic Neighbor Embedding) was applied to the model inputs. The high-dimensional multimodal time series is passed directly to the network, which performs implicit dimensionality reduction through sequential convolution, pooling, and adaptive average pooling layers before final classification. This end-to-end approach avoids the manual loss of fine-grained temporal structure that can occur when compressing raw windows into hand-crafted summary statistics.

3.5 Training and Validation Strategy

To ensure robust evaluation and prevent temporal data leakage, a subject-aware, class-stratified, chronological 60/20/20 train/validation/test split was applied independently within each subject’s recording timeline. The first 60% of each subject’s windows chronologically formed the training set, the next 20% formed the validation set, and the final 20% served as the held-out test set. The validation set was strictly utilized for early stopping and hyperparameter tuning (such as validating the selection of per-recording z-score over per-window normalization). Class imbalance within the training split was addressed using a combination of 5:1 interictal subsampling and positive class weighting in the cross-entropy loss function. To improve prediction robustness and mitigate random initialization effects, an ensemble of five independently trained models was utilized, with predicted probabilities averaged across runs during inference.

3.6 AI and Machine Learning Models

The primary model developed in this study is a Convolutional Neural Network (CNN), termed SeizureCNN, which is optimized for automated feature extraction directly from sequential physiological waveforms. The SeizureCNN architecture consists of three 1D convolutional blocks—each incorporating batch normalization, ELU activations, max pooling, and dropout—followed by an adaptive average pooling layer and a fully connected classifier outputting logits for the target classes (pre-ictal vs. interictal).

To benchmark the primary deep learning network, a Random Forest classifier serves as the baseline

model. The Random Forest baseline operates on the handcrafted log band-power spectral features, providing an interpretable point of comparison to evaluate whether end-to-end deep representation learning extracts superior predictive patterns from raw ambulatory waveforms. The overall research workflow is illustrated in Figure 1.

3.7 Research Pipeline:

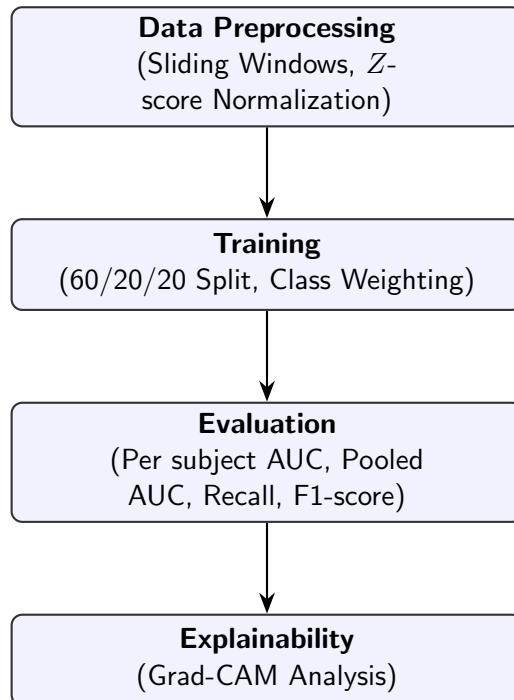


Figure 1: EEG/ECG Classification Pipeline Workflow

3.8 Tools, Programming Languages, and Frameworks

The project utilizes Python as the primary programming language for all data processing and model training. PyTorch serves as the core deep learning framework to build and train the primary CNN architecture, while scikit-learn is utilized to execute the Random Forest baseline and compute evaluation metrics. An overview of the core software libraries used in this study is provided in Table 2.

Table 2: **Tools, Programming Languages, and Frameworks**

Library	Version	Primary Function
python	3.11+	Primary development language and runtime framework.
argparse	standard	Parses command-line arguments and options for Python scripts.
re	standard	Provides regular expression matching and string manipulation.
dataclasses	standard	Generates boilerplate code (init, repr, etc.) for data-holding classes.
pathlib	standard	Object-oriented filesystem path handling and file I/O.
matplotlib	3.10.9	2D plotting library for creating static, animated, and interactive visualisations.
mne	1.12.1	Processes and analyses EEG, MEG, and other neurophysiological data.
numpy	2.4.6	Fundamental array computing: n-dimensional arrays, linear algebra, and math operations.
torch	2.12.0	Deep learning framework with GPU-accelerated tensor computation and neural network building.
torch.nn	2.12.0	Defines neural network layers, loss functions, and model containers.
torch.utils.data	2.12.0	Provides DataLoader and Dataset utilities for efficient batched data loading.
sklearn.metrics	1.9.0	Computes model evaluation metrics: confusion matrix, precision, recall, F1-score.
tqdm	4.68.2	Displays fast, extensible progress bars for loops and iterables.
typing	standard	Provides type hint support (List, Tuple, Dict, Optional, etc.) for static analysis.
captum	0.9.0	Model interpretability toolkit; provides Grad-CAM (LayerGradCam) to attribute predictions to input regions.

3.9 Evaluation and Explainability

runs shorter than a minimum duration to reduce false positives and improve clinical interpretability.

Model performance was comprehensively evaluated using a combination of threshold-independent and threshold-dependent metrics. A temporal post-processing step was implemented during inference, which utilized a minimum run-length filter of 2 consecutive windows to remove isolated

positive window spikes, thereby suppressing spurious false alarms and optimizing clinical interpretability. To inspect the internal representation of the primary neural network, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to the final convolutional layer of the 1D CNN, mapping back to the raw input sequence to visualize which localized temporal regions contributed most heavily to pre-ictal alarm triggers.

3.10 Performance Evaluation Metrics

The prediction system’s performance was evaluated using multiple complementary metrics chosen specifically to handle the unique challenges of seizure prediction in wearable, highly imbalanced physiological streams. Rather than relying strictly on a single pooled dataset score, the evaluation architecture splits metrics to analyze both global performance and honest, patient-specific utility.

The primary headline metric defined for this study is the per-subject Area Under the ROC Curve (AUC-ROC). Because the absolute output probability scales of a global neural network are uncalibrated and vary drastically between individual patients, pooling all windows across subjects before computing an evaluation metric creates a ranking artifact that masks true within-patient predictive power. The per-subject AUC measures the model’s capacity to correctly rank pre-ictal windows above baseline interictal windows within each individual patient’s separate timeline, serving as the most transparent and clinically relevant reflection of performance. For completeness, global/pooled AUC-ROC and Area Under the Precision-Recall Curve (AUC-PR) are also reported.

To assess threshold-dependent diagnostic reliability, Precision, Recall (Sensitivity), Specificity, and the F1-Score are computed, alongside a complete Confusion Matrix breakdown. Recall reflects the model’s sensitivity in flagging oncoming pre-ictal states, which translates directly to patient safety by ensuring warning flags are successfully raised before a seizure strikes. Conversely, Specificity tracks the identification of true interictal windows, which is essential to minimize false positive alarms that degrade a patient’s quality of life via unnecessary stress. Precision and the unified F1-Score ensure that the critical trade-off between missing a pre-ictal window and triggering an excessive alarm rate is evenly balanced under severe class imbalance. The selected evaluation metrics and their direct contextual justifications are summarized in Table 3.

Table 3: **Performance, Reliability and Robustness**

Name	Primary Function	Justification
Headline Patient Metric		
Per-Subject AUC-ROC	Measures the model’s capacity to rank pre-ictal windows higher than interictal windows within an individual subject.	Primary Headline Metric: Provides a fair, honest evaluation of within-patient ranking ability without being distorted by cross-patient pooling scale artifacts.
Global Metrics		
Accuracy	Measures the total proportion of correct window-level predictions across all samples.	Provides a broad, general indicator of classification correctness across the entire combined sample pool.
Pooled AUC-ROC / AUC-PR	Measures overall threshold-independent class discrimination on the combined dataset.	Critical for Patient Safety: Ensures oncoming pre-ictal states are caught early, allowing for timely alarms and clinical interventions.
F1-Score	Computes the harmonic mean of Precision and Recall into a single unified value.	Essential for clinical data streams where baseline interictal windows are vastly more common than pre-ictal windows.
Diagnostic Reliability		
Sensitivity (Recall)	Measures the proportion of true pre-ictal windows correctly flagged by the model.	Essential for clinical data where one outcome, such as seizure freedom, is more common.
Area Under the ROC Curve (AUC)	Measures the model’s overall diagnostic value.	Essential for clinical data streams where baseline interictal windows are vastly more common than pre-ictal windows.

4 Experiments and Results

This chapter presents the experimental setup and results of the seizure prediction system developed in this project. Two parallel pipelines are evaluated: one using only wearable EEG signals and one adding a single ECG channel as an additional input. Both are compared against a RandomForest baseline built on handcrafted frequency-band features. The central question addressed is whether ECG contributes meaningful predictive signal beyond EEG alone, evaluated on the held-out test set of 60 patients from the SeizeIT2 dataset using a combination of threshold-independent and

threshold-dependent performance metrics.

4.1 Experimental Setup

4.1.1 Tools and Libraries

All experiments were implemented in Python. Table 4 lists the key libraries used and their role within the project.

Table 4: Libraries and their purpose

Library	Version	Purpose
PyTorch	2.12.0	CNN model definition, training loop, GPU inference
MNE-Python	1.12.1	EDF file loading, notch filtering
scikit-learn	1.9.0	RandomForest baseline, evaluation metrics
NumPy	2.4.6	Window array construction and storage (.npz)
Captum	0.9.0	Grad-CAM explainability
Matplotlib / Seaborn	3.10.9	Result visualisation

Training was accelerated using PyTorch’s Automatic Mixed Precision (AMP) on a CUDA-capable GPU, reducing training time by approximately 2x compared to full-precision training.

4.1.2 Model Development

The codebase was structured into two parallel pipelines sharing a common core:

- **EEG-only** - using two wearable EEG channels as input.
- **EEG+ECG** - adding one ECG channel as a third input channel.

Each pipeline consisted of three stages executed sequentially: preprocessing (windowing and labelling), training, and evaluation. Shared logic — the CNN architecture, training loop, and evaluation routines — was centralised in common modules to ensure both pipelines were treated identically and results remained directly comparable.

Hyperparameters were first tuned on the validation set across multiple runs, varying the pre-ictal horizon, normalisation scheme, and learning rate. Once a configuration was selected, the model was retrained from scratch and evaluated once on the held-out test set. To assess robustness against random initialisation, training was repeated with five different random seeds (42-46), and a 5-model soft-voting ensemble was constructed as the primary result.

4.1.3 Validation Strategy

A within-subject chronological 60/20/20 split was applied independently per patient. The first 60% of each subject’s windows formed the training set, the next 20% the validation set, and the final 20% the held-out test set. This chronological ordering was enforced deliberately: splitting at random would allow the model to use signal from a future time point to predict an earlier one, constituting temporal data leakage. By respecting the recording timeline, the evaluation reflects how the model would perform prospectively.

Stratification by class label was not applied to the split. Pre-ictal windows occur in fixed temporal clusters immediately before seizure onsets and cannot be redistributed across splits without breaking chronological order. Class imbalance was therefore addressed through two separate mechanisms: interictal subsampling at a 5:1 ratio during preprocessing, and inverse-frequency class weighting in the training loss function.

No cross-patient splitting or k-fold cross-validation was used. Each subject’s data remained within its own split to prevent a model trained on one patient’s seizures from being evaluated on data that implicitly contains that patient’s patterns.

4.1.4 Baseline Comparison

A RandomForest classifier was used as a baseline to determine whether a CNN learning directly from raw waveforms outperforms handcrafted feature engineering. The baseline extracted log band-power in five standard EEG frequency bands (delta: 0.5-4 Hz, theta: 4-8 Hz, alpha: 8-13 Hz, beta: 13-30 Hz, gamma: 30-45 Hz) per channel, yielding a 10-feature vector per window for the EEG-only condition. The RandomForest was trained on the same training split and evaluated on the identical test set, ensuring a fair comparison.

4.2 Training and Evaluation

4.2.1 Training Procedure

The CNN was trained using the Adam optimiser with an initial learning rate of 1×10^{-3} and a cosine annealing learning rate schedule, which gradually reduces the learning rate over training epochs to improve convergence. Training ran for a maximum of 30 epochs with a batch size of 128.

Because the dataset is severely class-imbalanced - pre-ictal windows make up approximately 17% of the subsampled training set - a weighted cross-entropy loss was applied. The weight of the positive (pre-ictal) class was set to the ratio of interictal to pre-ictal samples, penalising missed pre-ictal predictions more heavily than false alarms.

Early stopping was applied with a patience of 5 epochs, monitoring validation AUC-ROC. If no improvement was observed for 5 consecutive epochs, training halted and the checkpoint with the

highest validation AUC was restored. This prevents overfitting to the training set while keeping the test set strictly unseen.

The full training procedure was repeated five times with different random seeds, and a soft-voting ensemble was constructed by averaging the predicted class probabilities across all five models before applying the decision threshold.

After training, the saved model checkpoint was loaded and applied once to the held-out test set. Threshold-dependent metrics (precision, recall, F1) were computed on the raw predicted probabilities, with isolated single-window positive predictions removed using a minimum run-length filter of 2 consecutive windows. Additionally, a 120-second centered moving-average was applied separately to evaluate its effect on AUC-ROC.

4.2.2 Hyperparameter Selection

Hyperparameters were divided into two categories: those set by design based on the literature and data properties, and those tuned empirically on the validation set.

Table 5: Hyperparameters fixed by design

Hyperparameter	Value	Rationale
Window size	2 s	Standard in EEG seizure prediction literature.
Step size	1 s	50% overlap balances data volume and redundancy.
Sampling rate	256 Hz	Fixed by the SeizeIT2 dataset.
Notch filter	50 Hz	European power-line frequency.
Batch size	128	Standard for time-series CNN training.
Dropout	0.25 / 0.50	Conservative regularisation values.
Interictal ratio	5:1	Balances class distribution while retaining data volume.

Table 6: Hyperparameters tuned on the validation set

Hyperparameter	Options tested	Selected value
Pre-ictal horizon	10 min, 20 min, 30 min	10 min
Input normalisation	Per-window z-score, per-recording z-score	Per-recording
Temporal smoothing	0 s, 30 s, 60 s, 120 s	120 s

The pre-ictal horizon of 10 minutes was selected over longer horizons of 20 and 30 minutes on both signal quality and clinical grounds. A longer horizon introduces early pre-ictal windows that are physiologically difficult to distinguish from interictal activity, adding noise to the positive class without a corresponding increase in detectable signal. A 10-minute warning window also remains clinically actionable for seizure intervention, whereas a 30-minute horizon may include periods too distant from onset to carry consistent precursor patterns.

Per-recording z-score normalisation was selected over per-window normalisation based on validation AUC (0.554 vs. 0.535). Per-window normalisation rescales every window independently, discarding amplitude variation across the recording that may carry pre-ictal information. Per-recording normalisation preserves these relative amplitude differences while still correcting for long-term drift between recordings.

Temporal smoothing of 120 seconds was selected as the best-performing option across all evaluation runs, consistently outperforming shorter windows of 30 and 60 seconds. A 120-second moving average corresponds to 120 consecutive 2-second windows, representing 2 minutes of sustained signal - a duration that aligns with the gradual nature of pre-ictal state transitions rather than momentary EEG fluctuations

4.2.3 Evaluation Metrics

Given the class imbalance in the dataset, a combination of threshold-dependent and threshold-independent metrics was used to capture different aspects of model performance.

- **Threshold-independent metrics** (do not depend on the decision boundary):
 - **AUC-ROC:** Area under the receiver operating characteristic curve. Measures how well the model ranks pre-ictal windows above interictal windows regardless of the chosen threshold. A value of 0.5 corresponds to random chance; 1.0 to perfect separation.
 - **AUC-PR:** Area under the precision-recall curve. More informative than AUC-ROC under class imbalance, as it is sensitive to false positive behaviour.
 - **Per-subject AUC:** AUC computed independently per patient, then averaged across all 60 subjects. This is the primary reported metric, because pooling all subjects into a single ranking is distorted by differences in predicted probability scale between patients.
- **Threshold-dependent metrics** (computed at decision threshold 0.5):
 - **Precision:** Fraction of positive predictions that are correct ($\frac{TP}{TP+FP}$); reflects the false alarm rate.
 - **Recall (Sensitivity):** Fraction of actual pre-ictal windows correctly identified ($\frac{TP}{TP+FN}$); reflects the missed seizure rate.
 - **F1-score:** Harmonic mean of precision and recall; the primary threshold-dependent metric.
 - **Confusion matrix:** TP, TN, FP, FN counts; provides a complete breakdown of classification outcomes.
- **Overfitting indicator:**
 - **Train F1 vs. Test F1:** F1 computed on a subsample of the training set and compared to the test F1. A large gap indicates the model memorised training data rather than generalising.

4.3 Results

4.3.1 Overall Performance

Table 2 presents the full test-set performance of all three models. The CNN results are shown for both the 5-model ensemble (primary) and as mean $\pm$ standard deviation across five individual seeds to indicate robustness.

Table 7: Test-set performance — all models

Model	AUC-ROC	AUC-ROC (sm.)	Per-subj. AUC	Precision	Recall	F1	AUC-PR
CNN — EEG-only (ens.)	0.586	0.609	0.497	0.203	0.655	0.310	0.209
CNN — EEG-only (mean $\pm$ SD)	0.580 ± 0.005	0.603 ± 0.006	0.503 ± 0.021	0.204 ± 0.011	0.635 ± 0.068	0.307 ± 0.003	—
CNN — EEG+ECG (ens.)	0.576	0.592	0.519	0.199	0.531	0.289	0.211
CNN — EEG+ECG (mean $\pm$ SD)	0.572 ± 0.008	0.592 ± 0.009	0.512 ± 0.015	0.202 ± 0.003	0.511 ± 0.045	0.291 ± 0.008	—
RandomForest (baseline)	0.566	—	0.525	0.221	0.130	0.164	0.208

Note: Smoothed AUC = AUC after 120-second moving-average post-processing. AUC-ROC = 0.5 corresponds to random chance.

4.3.2 Confusion Matrices

Tables 8-10 show the confusion matrices for all three models on the test set ($n = 106,340$ windows total, of which 17,977 are pre-ictal).

Table 8: Confusion matrix — CNN EEG-only ensemble

		Predicted	
		Interictal	Pre-ictal
Actual	Interictal	42,150 (TN)	46,213 (FP)
	Pre-ictal	6,207 (FN)	11,770 (TP)

Table 9: Confusion matrix — CNN EEG+ECG ensemble

		Predicted	
		Interictal	Pre-ictal
Actual	Interictal	49,921 (TN)	38,442 (FP)
	Pre-ictal	8,432 (FN)	9,545 (TP)

Table 10: Confusion matrix — RandomForest baseline

		Predicted	
		Interictal	Pre-ictal
Actual	Interictal	80,117 (TN)	8,246 (FP)
	Pre-ictal	15,642 (FN)	2,335 (TP)

4.3.3 ROC Cruves

(Insert figure: ROC curves for EEG-only CNN, EEG+ECG CNN, and RandomForest plotted in one figure, with the diagonal chance line)

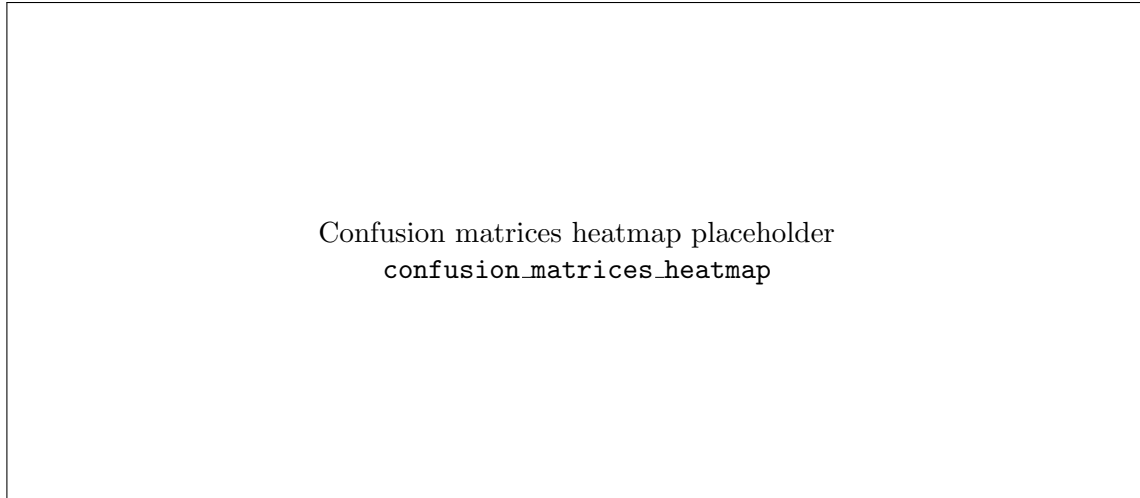


Figure 2: Confusion matrices visualised as heatmaps for all three models side-by-side.

The ROC curves confirm the ranking seen in Table 2. The EEG-only CNN sits consistently above EEG+ECG and RandomForest across the majority of operating points, though the separation between models is small. All three curves remain close to the diagonal, reflecting the overall difficulty of the task.

4.3.4 Commentary: Which Model Performed Better, and Why

CNN outperforms RandomForest. The EEG-only CNN ensemble achieves nearly twice the F1 score of the RandomForest (0.310 vs. 0.164) and substantially higher recall (0.655 vs. 0.130). This demonstrates that learning temporal features directly from the raw waveform is more effective than summarising the signal as frequency-band power values. The CNN can capture short-duration transients and morphological patterns within a 2-second window that are invisible to band-power features.

The RandomForest trades recall for precision: it issues far fewer false alarms (8,246 FP vs. 46,213 for the CNN) but misses the majority of pre-ictal windows (15,642 FN). In seizure prediction, missing a pre-ictal period is the more critical failure - this behaviour makes the RandomForest less suited to the task despite its cleaner confusion matrix.

The effect of ECG is metric-dependent. Whether ECG helps depends on which metric is considered. On the threshold-dependent side, EEG-only achieves a higher F1 (0.310 vs. 0.289): the EEG+ECG model issues fewer false alarms (38,442 vs. 46,213 FP) but misses substantially more pre-ictal windows (8,432 vs. 6,207 FN), and its larger train-test F1 gap (+0.112 vs. +0.016 for EEG-only, reflecting a training F1 of 0.401 vs. 0.326) shows that the extra channel adds capacity that the model partly overfits to the training recordings. On the primary, threshold-independent metric, however, the ranking reverses: EEG+ECG attains a higher per-subject AUC than EEG-only (0.519 vs. 0.497 for the ensemble; 0.512 vs. 0.503 across seeds), a direction independently reproduced by the RandomForest baseline (0.554 vs. 0.525). The ECG channel therefore does carry genuine within-patient pre-ictal signal, but this signal is not exploitable by a single global decision

threshold — a tension examined in Section ??

Where the CNN succeeds: The EEG-only CNN correctly identifies 65.5% of all pre-ictal windows, demonstrating that wearable EEG contains detectable pre-ictal signal for a meaningful portion of the data. The smoothed AUC-ROC of 0.609 confirms that when predictions are aggregated over time, the model captures a genuine statistical trend in pre-ictal periods rather than producing purely random outputs.

Where the CNN fails: Precision remains low across all models (0.20), meaning approximately 4 out of every 5 positive predictions are false alarms. This is partly expected given the class distribution — even after 5:1 subsampling, interictal windows dominate the test set, giving the model many opportunities to generate false positives. More fundamentally, the per-subject AUC of 0.497 for the EEG-only ensemble indicates that when evaluated within individual patients, the model’s probability rankings are only marginally better than random on average. The global AUC-ROC of 0.586 is therefore partly inflated by between-subject structure rather than genuine within-patient seizure prediction.

4.4 Unexpected Results

4.4.1 ECG Provides Only a Marginal, Patient-Dependent Benefit

ECG is a recognised biomarker of autonomic nervous system activity, and autonomic changes — including heart rate variability shifts — have been reported as precursors of seizure onset. It was therefore expected that ECG would broadly complement the EEG signal and improve classification. The actual effect was more subtle: averaged over all patients, ECG produced only a small improvement on the primary per-subject AUC (0.519 vs. 0.497) and no improvement on the threshold-dependent F1 (0.289 vs. 0.310), while increasing the train–test gap (+0.112 vs. +0.016).

The unexpected aspect is not a uniform null effect but the *heterogeneity* behind the average. ECG raised the per-subject AUC substantially for some patients while lowering it for others, so the small positive mean is the residue of opposing per-patient effects that largely cancel. Rather than indicating that ECG carries no useful signal, this suggests that its value is patient-specific and that a single global model — trained and thresholded across all patients — cannot exploit it consistently. The implications of this finding for personalised modelling are discussed in Section ??.

4.4.2 Global AUC Underestimates Per-subject Performance

The pooled AUC-ROC of 0.586 appears modest, yet the per-subject AUC averages 0.497 — seemingly contradicting each other. This is explained by a pooling artefact. When all 60 subjects’ predicted probabilities are combined into a single ranking, subjects whose predictions are systematically higher or lower than average dominate the ranking, regardless of within-patient discriminability. Because the CNN’s absolute probability outputs are not calibrated to a common scale across patients, the pooled AUC reflects between-subject differences in output level rather than the model’s actual ability to predict seizures within a patient.

Per-subject AUC is therefore the more honest metric: it measures within-patient ranking ability, which is the clinically relevant question.

4.4.3 Large Spread in Per-subject Performance

A striking finding is the range of per-subject AUC scores, from 0.111 (sub-057) to 0.835 (sub-018), with roughly one third of subjects scoring below chance level ($\text{AUC} < 0.5$). This degree of variability was not anticipated from a model trained on data from all 60 patients.

Two factors likely contribute to this. First, the biological variability in seizure precursors across patients means that not all patients exhibit consistent, detectable pre-ictal EEG changes at the wearable electrode positions used in SeizeIT2. Second, the number of seizures per patient in the test set is small - many subjects have only one seizure contributing pre-ictal windows, making the per-subject AUC estimate highly sensitive to a single prediction sequence. For these subjects, it is difficult to determine whether a low AUC reflects a genuine absence of pre-ictal signal or simply insufficient test data.

Table 11: Distribution of per-subject AUC - EEG-only CNN ensemble

AUC range	Number of subjects
≥ 0.70 (strong)	11
$0.50 - 0.69$ (above chance)	25
< 0.50 (below chance)	24

4.4.4 Temporal Smoothing Improves AUC Consistently

It was initially expected that the CNN’s per-window predictions would already capture temporal structure, given that convolution operates over the full 512-sample window. However, applying a 120-second moving average to the predicted probabilities consistently improved AUC-ROC by approximately 0.023 across all seeds and both pipelines. This suggests that the model produces noisy per-window predictions - high-confidence pre-ictal scores interspersed with interictal scores even within a genuinely pre-ictal period - and that genuine pre-ictal periods are better characterised by a sustained elevated probability trend than by any individual window. The smoothing step effectively acts as a second-stage temporal classifier on top of the raw CNN output.

5 Discussion and Evaluation

5.1 Interpreting the Results Against the Research Question

The results give a nuanced rather than a binary answer. On the primary, threshold-independent metric defined for this study — the per-subject AUC — adding ECG produced a small but consistent improvement: from 0.497 to 0.519 for the ensemble, and from 0.503 ± 0.021 to 0.512 ± 0.015

across the five seeds (Table 7). The same direction held after temporal smoothing and, importantly, was reproduced independently by the RandomForest baseline, whose per-subject AUC also rose with ECG (0.525 to 0.554). That two unrelated model families agree on the direction indicates the cardiac channel carries genuine, if weak, pre-ictal information rather than a single-model artefact.

At the same time, ECG did *not* improve the pooled AUC-ROC (0.576 vs. 0.586) or the F1 score at the 0.5 threshold (0.289 vs. 0.310), and it increased the train-test F1 gap (+0.112 vs. +0.016). The answer to the research question is therefore conditional: **ECG adds a marginal amount of within-patient discriminative signal, but this benefit does not translate into better global, threshold-based decision performance.** Reconciling these two observations — rather than reading them as a contradiction — is the central interpretive task of this chapter, and it points directly to patient heterogeneity and the limits of a single global model.

5.2 Per-subject AUC as the Honest Metric

The divergence between the per-subject AUC and the pooled AUC is the most important methodological observation of this work. Because each patient has a different baseline physiology and electrode contact, the network’s output probabilities are calibrated differently per patient: a probability that is “high” for one subject may be “low” for another. Pooling all subjects into one ROC curve mixes these inconsistent probability scales, so the between-patient offset dominates the within-patient ranking that actually reflects pre-ictal detection. The pooled AUC consequently collapses toward chance (≈ 0.58) and is largely insensitive to real improvements — which is precisely why the ECG benefit is invisible there but visible in the per-subject AUC. For the same reason, the F1 at a single global threshold is an unreliable summary: one fixed cut-off cannot be appropriate for patients whose probability distributions differ so widely. We therefore treat the per-subject AUC as the primary outcome and report pooled AUC and F1 only for completeness.

5.3 The Core Finding: Patient Heterogeneity and the Limits of a Global Model

The dominant feature of the results is not the small average ECG effect but the extreme **heterogeneity across patients**. Per-subject AUC ranged from 0.111 to 0.835, with roughly one third of subjects scoring below chance (Table 11). Critically, the effect of ECG was *not uniform*: for some patients it raised the per-subject AUC substantially (e.g. ⟨sub-040⟩, from ⟨0.05⟩ to ⟨0.85⟩), whereas for others it clearly reduced it (e.g. ⟨sub-028⟩, from ⟨0.90⟩ to ⟨0.72⟩). The small positive average therefore emerges only after these opposing per-patient effects partially cancel (Figure ??).

This pattern carries the main message of the study. A single model trained across all patients, applied with a single decision threshold, implicitly assumes a shared pre-ictal signature — yet the data show that pre-ictal precursors, and the degree to which ECG contributes to them, are strongly patient-specific. A global “one-model-fits-all” approach therefore averages away exactly the personalised signal that ECG provides for the patients who benefit, which is why the per-subject gain does not survive into the pooled and threshold-based metrics. The results are thus better read not as “ECG does not help” but as **evidence that the value of ECG is patient-dependent,**

and that personalised, per-patient modelling — including per-patient feature selection (EEG vs. EEG+ECG) and per-patient decision thresholds — is a more promising route to clinically useful seizure prediction than a single shared model. This reframes the modest average performance as a property of the modelling strategy rather than a verdict on the signals themselves.

5.4 CNN versus the RandomForest Baseline

The baseline comparison qualifies the role of deep learning here. On the threshold-dependent metrics the CNN was clearly stronger — nearly double the F1 (0.310 vs. 0.164) and far higher recall (0.655 vs. 0.130) — because the RandomForest, operating on five band-power summaries, defaults to a high-precision/low-recall regime that misses most pre-ictal windows, the more costly error in a warning system. On the primary per-subject AUC, however, the shallow model was on par with, and nominally slightly above, the CNN (0.525 / 0.554 vs. 0.497 / 0.519). Two implications follow. First, the corroboration of the ECG direction by an interpretable model strengthens confidence in the finding. Second, the fact that a tree model on five spectral bands matches a CNN on the raw waveform for within-patient ranking suggests that, at this dataset size and signal quality, the **ceiling is set by the data rather than by model capacity** — additional representational power does not extract additional generalisable signal.

5.5 Trade-offs and Practical Significance

Recall versus false alarms. The threshold sweep exposes the core operating trade-off: at low thresholds the model reaches very high recall (> 0.95) but precision near the positive base rate (≈ 0.17), i.e. a heavy false-alarm burden; raising the threshold improves precision only marginally while recall collapses. For a prediction (warning) system, high recall is preferable, but the corresponding false-alarm rate at this performance level would be impractical for real-world use — a concrete illustration of why the modest AUC matters clinically.

Performance versus explainability. Interpretability was retained alongside the deep model: the RandomForest provides global feature-importance rankings — in which the ECG bands ranked highest for the EEG+ECG condition, a direct confirmation that the model uses the cardiac signal — and Grad-CAM provides per-window saliency for the CNN. Because the CNN did not outperform the interpretable baseline on the primary metric, the usual “accuracy-for-opacity” trade-off does not pay off in this setting; the interpretable model is arguably the more attractive option, which is itself a useful practical finding.

Sensor cost versus benefit. From a wearable-design perspective, the question is whether the added hardware, comfort, and processing cost of an ECG lead is justified by an average per-subject AUC gain of ≈ 0.02 . On these results the benefit is real but small and, as argued above, concentrated in a subset of patients — making ECG most defensible as part of a *personalised* configuration for the patients it demonstrably helps, rather than as a blanket addition for all.

5.6 Robustness, Generalisability, and Limitations

The multi-seed protocol supports the robustness of the comparison: the standard deviation of the per-subject AUC across seeds was small (< 0.021), so the EEG+ECG advantage is reproducible rather than an effect of initialisation. Temporal smoothing improved every configuration, confirming that a single 2-second window is a noisy estimate and that tracking the probability trend over time is both clinically realistic and methodologically sound.

Several limitations bound the conclusions:

- **Within-subject evaluation.** The chronological 60/20/20 split is per patient, so every subject appears in all three sets. This measures *personalised* prediction and does not demonstrate generalisation to unseen patients; a leave-subjects-out setting is expected to be substantially harder and was outside the scope of the primary evaluation.
- **Small, imbalanced per-subject test sets.** Many subjects contribute only one or two seizures, so their per-subject AUC is high-variance, and for some subjects a low score cannot be distinguished from simply insufficient pre-ictal test data.
- **Paired-comparison integrity.** The authoritative comparison was run on the subjects that have an ECG recording, so both feature sets are evaluated on identical recordings — the correct fair comparison, at the cost of a slightly smaller cohort.
- **Modest absolute performance and signal quality.** Two-channel behind-the-ear EEG and a single ambulatory ECG lead are inherently lower-SNR than clinical scalp montages, and the long, predominantly interictal recordings make this a genuinely hard problem.
- **Fixed labelling choices.** The 10-minute pre-ictal horizon, 10-minute post-ictal guard, and 5:1 interictal ratio were fixed; the operating point is conditioned on these definitions.

5.7 Comparison with Related Work

Seizure-prediction studies on clinical multi-channel scalp EEG routinely report markedly higher AUC and sensitivity (cite the related-work studies from Section 2). Our lower absolute numbers do not contradict that literature; they reflect a deliberately harder and more realistic setting — a small number of behind-the-ear channels from an ambulatory wearable, patient-independent labelling rules, and a strict pre-ictal/interictal *prediction* framing rather than ictal detection. Where prior work has investigated autonomic (ECG/HRV) precursors of seizures (cite), our finding of a small, patient-dependent ECG contribution is directionally consistent, while the controlled paired design isolates that contribution more cleanly than studies that vary several factors simultaneously. The contribution of this work relative to prior art is therefore not a new benchmark score but (i) a fair, reproducible, within-patient-honest quantification of how much a single ECG channel adds on a realistic wearable setup, and (ii) the evidence that this added value is patient-specific, motivating a shift from global to personalised modelling.

6 Conclusion and Future Work

6.1 Summary of Findings and Contributions

This project investigated whether adding a single ECG channel to wearable behind-the-ear EEG improves epileptic seizure *prediction*, using a 1D convolutional neural network on the SeizeIT2 dataset under a strictly controlled, paired comparison in which the two feature sets differed by nothing other than the ECG channel. A RandomForest classifier on handcrafted band-power features served as an interpretable baseline.

The central finding is nuanced rather than binary. On the primary, threshold-independent metric — the per-subject (within-patient) AUC — ECG produced a small but consistent improvement (0.497 to 0.519 for the ensemble), a direction independently reproduced by the RandomForest baseline (0.525 to 0.554). This gain did not, however, translate into better pooled AUC or threshold-based F1, and it was accompanied by a larger train-test gap. The explanation lies in the dominant characteristic of the results: **extreme heterogeneity across patients**. ECG raised the per-subject AUC substantially for some patients while lowering it for others, so the modest positive average is the residue of opposing per-patient effects. The work therefore concludes not that “ECG does not help”, but that the value of ECG is *patient-specific*, and that a single global “one-model-fits-all” model — trained and thresholded across all patients — averages away exactly the personalised signal that benefits the patients for whom ECG is informative.

The main contributions of this work are:

- A **fair, reproducible, paired comparison** of EEG-only versus EEG+ECG feature sets for seizure prediction on a realistic wearable dataset, isolating the contribution of the cardiac channel from all other factors.
- A demonstration that the **per-subject AUC is the honest evaluation metric** in this setting, and that pooled AUC and global-threshold F1 mask genuine within-patient effects because the model’s probability scale is not comparable across patients.
- Evidence that, at this dataset size and signal quality, a **shallow interpretable baseline matches the deep CNN** on the primary metric, indicating that the performance ceiling is set by the data rather than by model capacity.
- The key insight that **seizure prediction from wearable signals is inherently a personalised problem**, providing empirical motivation for per-patient modelling.

6.2 Reflection on the Body of Knowledge and Practice

Relative to prior seizure-prediction studies, which typically report high performance on multi-channel clinical scalp EEG, this work contributes a more realistic and more critical picture: on a small number of ambulatory behind-the-ear channels, under a strict pre-ictal/interictal prediction framing, average performance is modest and the added value of multimodal (EEG+ECG) fusion

is small and patient-dependent. The practical implication is twofold. First, multimodal sensing should be evaluated and, ultimately, deployed selectively — ECG is most defensible as part of a personalised configuration for the patients it demonstrably helps, rather than as a blanket addition for all. Second, evaluation practice matters: reporting only pooled metrics would have led to the misleading conclusion that ECG is useless, whereas the within-patient analysis reveals where the signal actually resides. These methodological points are arguably as valuable as the numerical results themselves.

6.3 Limitations Carried into Future Work

The conclusions are bounded by the within-subject evaluation design (which measures personalised prediction rather than generalisation to unseen patients), the small and imbalanced per-subject test sets (often only one or two seizures per patient), and the modest signal quality inherent to two-channel wearable EEG and a single ambulatory ECG lead. These limitations directly shape the recommendations below.

6.4 Future Work and Recommendations

- **Personalised modelling.** The strongest direction following from this work is a per-patient approach: per-patient feature selection (using EEG-only or EEG+ECG depending on which helps that individual), per-patient decision thresholds, and patient-specific fine-tuning or calibration of a shared base model. This directly targets the heterogeneity that limited the global model.
- **Cross-patient evaluation.** A leave-subjects-out protocol should be added to quantify how well the model generalises to entirely unseen patients — the harder and more clinically relevant deployment scenario not covered by the within-subject split here.
- **Richer cardiac and multimodal features.** Rather than feeding the raw ECG waveform, deriving explicit autonomic features (e.g. heart-rate variability metrics) may expose the pre-ictal signal more directly, and the additional SeizeIT2 modalities (EMG, movement) could be incorporated to test broader multimodal fusion.
- **Event-level and alarm-based evaluation.** Future studies should complement window-level metrics with clinically oriented measures such as sensitivity per seizure and false-alarm rate per hour, which more faithfully reflect how a warning system would be used.
- **Larger and more balanced cohorts.** Validating on additional patients and more seizures per patient would stabilise the per-subject estimates and allow the patient-specific ECG effect to be characterised — for instance, identifying clinical or signal-based features that predict *which* patients benefit from ECG.
- **Deployment considerations.** A real wearable system would further require on-device or low-latency inference, robustness to motion and electrode artefacts, and the temporal post-processing shown here (probability smoothing over time) to suppress spurious single-window alarms.

In summary, this work shows that a single ECG channel adds only a marginal and patient-dependent amount of predictive signal to wearable EEG for seizure prediction, and that the path to clinically useful performance lies less in adding sensors to a global model than in tailoring the model — its features and its thresholds — to the individual patient.

7 Availability of Data and Materials

7.1 GitHub

The source code developed for this project — including the preprocessing pipeline, model implementation, training scripts, and evaluation code — is publicly available at:

<https://github.com/Lijcoesh/capstone-project>

7.2 Dataset

This project uses the publicly available **SeizeIT2** dataset [35], a BIDS-formatted collection of wearable recordings (behind-the-ear EEG, ECG, EMG, and movement data) acquired with the Sensor Dot device in European Epilepsy Monitoring Units. The dataset is hosted on OpenNeuro (accession `ds005873`) and is released under the CC0 license, allowing free reuse.

To keep the repository lightweight, the large raw EDF signals are not redistributed; only the BIDS metadata and per-recording annotations are included. The full signals can be downloaded from OpenNeuro and placed under `data/raw/seizeit2/`, following the BIDS layout. Step-by-step instructions for reproducing the experiments are provided in the repository README.

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